7 - Sparse Coding Minimizing ℓ_0: Matching Pursuit

MMM for Image Processing

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March 23, 2026

1 The Setup — What is Sparse Coding?

- **Goal:** You have a signal $s \in \mathbb{R}^m$ that you want to represent as a linear combination of columns of a **dictionary** $D \in \mathbb{R}^{m \times N}$:

$$s = Dx = \sum_{i=1}^{N} x_i d_i \tag{1}$$

- **Redundancy Constraint:** The key constraint is $N > m$ — the dictionary has more columns (atoms) than rows (signal dimension). This is called **redundancy** (or an overcomplete dictionary).
- **The Problem:** Because $N > m$, the system $s = Dx$ is underdetermined. There are infinitely many coefficient vectors x that satisfy it. We want the **sparsest** one: a vector x with as many $x_i = 0$ as possible.

2 Why Not Just Minimize ℓ_2? (Ridge Regression)

The most natural initial approach is to solve problem (P_2^c):

$$\hat{x} = \arg\min_x \|x\|_2^2 \quad \text{s.t.} \quad s = Dx \tag{2}$$

Using Lagrange multipliers (or the pseudo-inverse), the unconstrained relaxation with regularization gives the **Ridge Regression** closed-form solution:

$$\hat{x} = (D^\top D + \lambda I_N)^{-1} D^\top s, \quad \lambda > 0 \tag{3}$$

3 Why is $D^\top D$ Not Directly Invertible?

- $D^\top D$ is an $N \times N$ matrix.
- However, D is $m \times N$ with $N > m$.
- By the properties of rank:

$$\operatorname{rank}(D^\top D) \leq \min(\operatorname{rank}(D^\top), \operatorname{rank}(D)) \leq m < N \tag{4}$$

- **Conclusion:** $D^\top D$ has rank at most m, but it is an $N \times N$ matrix. Therefore, it is **not full rank** and cannot be inverted directly.

4 Proof that $D^\top D$ is Symmetric Positive Semi-Definite

Let $A = D^\top D$. We must prove two properties:

- **Symmetry:**
$$A^\top = (D^\top D)^\top = D^\top (D^\top)^\top = D^\top D = A \quad \checkmark \tag{5}$$

- **Positive Semi-Definiteness (SPSD):** For any arbitrary vector $x \in \mathbb{R}^N$, let $y = Dx$. Then:
$$x^\top D^\top D x = (Dx)^\top (Dx) = y^\top y = \|y\|_2^2 \geq 0 \quad \forall x \quad \checkmark \tag{6}$$

Because A is symmetric, it admits an orthogonal eigendecomposition:

$$A = U^\top \Lambda U, \quad UU^\top = I_N, \quad \Lambda = \mathrm{diag}(\lambda_i), \quad \lambda_i \geq 0 \tag{7}$$

The eigenvalues are non-negative (but not necessarily strictly positive — hence *semi-definite*, not definite).

5 How Adding λI_N Makes it Invertible

Substitute $D^\top D = U^\top \Lambda U$ into the Ridge Regression matrix:

$$D^\top D + \lambda I_N = U^\top \Lambda U + \lambda U^\top U = U^\top (\Lambda + \lambda I_N) U \tag{8}$$

- The diagonal entries of $(\Lambda + \lambda I_N)$ are $\lambda_i + \lambda$.
- Since $\lambda_i \geq 0$ and $\lambda > 0$, we have:
$$\lambda_i + \lambda \geq \lambda > 0 \quad \forall i \tag{9}$$

- **Result:** The diagonal matrix $(\Lambda + \lambda I_N)$ is **strictly positive diagonal**, making it invertible. Since U is orthogonal (and thus invertible), the entire product is invertible.
- **Intuition:** Adding λI_N "lifts" all eigenvalues by λ, ensuring none are zero.
- **The Fatal Flaw:** Ridge regression gives a valid solution, but it distributes signal energy across *all* N coefficients. We want many coefficients to be exactly **zero**, but ℓ_2 minimization encourages small-but-nonzero values everywhere. It is **not sparse**.

6 Norms — Definitions

A **norm** $\|\cdot\| : \mathbb{R}^N \to \mathbb{R}$ must satisfy three fundamental axioms:

Property	Formula	Meaning
Positive definite	$\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$	Zero only at the origin
Homogeneity	$\|\alpha x\| = \|\alpha\|\|x\|$	Scales linearly
Triangle inequality	$\|x + y\| \leq \|x\| + \|y\|$	Convexity condition

7 The ℓ_p Family and Sparsity-Promoting Norms

The ℓ_p norm for $p \geq 1$ is defined as:

$$\|x\|_p = \left(\sum_{i=1}^{N} |x_i|^p \right)^{1/p} \tag{10}$$

Special Cases:

- $p = 2$ **(Euclidean):** $\|x\|_2 = \sqrt{\sum x_i^2} \to$ Circular ball.
- $p = 1$ **(Manhattan):** $\|x\|_1 = \sum |x_i| \to$ Diamond-shaped ball.
- $p = 1/2$**:** $\|x\|_{1/2} = (\sum \sqrt{|x_i|})^2 \to$ Star-shaped ball. (*Not a true norm* as it fails the triangle inequality, but useful as a penalty).
- $p \to 0$ **(ℓ_0):** $\|x\|_0 = \#\{i : x_i \neq 0\} \to$ Counts non-zero entries.

Why does a smaller p promote sparsity? Look at the geometry. As p decreases from $2 \to 1 \to 0.5$, the unit ball develops sharper "corners" along the axes. When you intersect this ball with the linear constraint $s = Dx$ (a hyperplane), the first point of contact is much more likely to occur exactly at a corner on an axis — meaning most $x_i = 0$.

Cauchy-Schwartz Inequality (Fundamental Tool):

$$\langle x, y \rangle \leq \|x\|_2 \|y\|_2 \tag{11}$$

8 The ℓ_0 "Norm" — Why it is Problematic

The ℓ_0 count is defined as:

$$\|x\|_0 = \#\{i : x_i \neq 0\} \tag{12}$$

Why it is NOT a true norm: It fails the homogeneity axiom.

$$\left\| \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\|_0 = 1, \quad \text{but} \quad \left\| 7 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\|_0 = 1 \neq 7 \cdot 1 \tag{13}$$

Scaling a vector by 7 does not change the count of its non-zero elements. Furthermore, ℓ_0 is:

- **Discrete:** It only takes integer values.
- **Discontinuous:** It jumps abruptly at zero; no derivative exists.
- **Conclusion:** No gradient-based optimization is possible!

9 Dictionary Representation & Geometric Sparsity

Assume a dictionary $D = [d_1 \mid d_2 \mid d_3 \mid d_4]$ (e.g., $m = 3, N = 4$).

$$s = \sum_{i=1}^{4} x_i d_i \tag{14}$$

What does $\|x\|_0 = k$ mean geometrically? It means the signal s lives strictly in the span of exactly k atoms of D.

- **$\|x\|_0 = 1$:** The signal lies on one of the 4 lines spanned by a single d_i.
- **$\|x\|_0 = 2$:** The signal lies in one of the $\binom{4}{2} = 6$ planes spanned by a pair (d_i, d_j).

The solution space for a sparse representation is a **union of subspaces**, not a single convex subspace. The sparsity question asks: *Which k-dimensional subspace does my signal s belong to?*

10 The (P_0^c) Problem and Why Brute Force Fails

The true optimization problem we want to solve is (P_0^c):

$$(P_0^c) \quad \hat{x} = \arg\min_{x \in \mathbb{R}^N} \|x\|_0 \quad \text{s.t.} \quad Dx = s \tag{15}$$

Note on Notation: The superscript c stands for *constrained* (equality constraint $Dx = s$), distinguishing it from an unconstrained Lagrangian form.

No closed-form solution exists. A **brute-force** approach would be to try all combinations of atoms, starting from $\|x\|_0 = 1$:

Sparsity Level	Combinations to Try	Sub-problem Size
$\|x\|_0 = 1$	$\binom{N}{1} = N$	$m \times 1$ system
$\|x\|_0 = 2$	$\binom{N}{2}$	$m \times 2$ system
$\|x\|_0 = k$	$\binom{N}{k}$	$m \times k$ system

For each combination, you form the sub-dictionary D_k and solve $D_k x = s$. **Why this is catastrophic:** Let $m = 500$, $N = 2000$, and we seek sparsity $\|x\|_0 = 20$.

$$\binom{2000}{20} \approx 3.9 \times 10^{47} \text{ combinations} \tag{16}$$

This has $O(N^k)$ complexity. Even computing 10^9 operations/second, this would take longer than the age of the universe. **Conclusion:** We need smarter, greedy algorithms (like Matching Pursuit) to avoid exhaustive search.

11 Deep Dive: The Mathematics of Atom Selection in MP

11.1 The Big Picture Aim

- We have a signal s and a dictionary of atoms $d_1, d_2, \ldots, d_N$.

- We want to pick **one atom** to add to our solution — specifically the atom that best explains s (or what remains of s so far).
- The question is: *Which atom d_j is "closest" to s?*

11.2 Idea 1 — For a Given Atom d_j, Find the Best Scaling z

- Atom d_j is just a direction (a fixed vector). We can scale it by any scalar number z. We want to find the z that makes $z \cdot d_j$ as close to s as possible. That means solving:

$$z^* = \arg\min_z \|s - z \cdot d_j\|_2^2 \tag{17}$$

- *Note:* z is a scalar and d_j is a vector. We are asking: "how far along the d_j direction should we travel to get closest to s?"
- This is a simple 1D calculus problem. Take the derivative with respect to z and set to zero:

$$\frac{d}{dz}\|s - z \cdot d_j\|_2^2 = -2\langle s - z \cdot d_j, d_j\rangle = 0 \tag{18}$$

- Solving gives the **projection formula**:

$$z^* = \frac{\langle s, d_j\rangle}{\|d_j\|_2^2} = \frac{s^\top d_j}{\|d_j\|_2^2} \tag{19}$$

  This is just the standard formula for projecting a vector onto a line. z^* tells us how much of d_j "fits inside" s.

11.3 Idea 2 — $\varepsilon(j)$: How Much is Left Unexplained?

- Once we place the best projection $z^* \cdot d_j$, there is still a gap between that projection and s. The squared length of that gap is the **residual error**:

$$\varepsilon(j) = \|s - z^* \cdot d_j\|_2^2 \tag{20}$$

- Expanding this algebraically by substituting z^* in:

$$\varepsilon(j) = \|s\|_2^2 - \frac{\langle s, d_j\rangle^2}{\|d_j\|_2^2} \tag{21}$$

- The second term represents "how much of s^2 we managed to explain using d_j." A large inner product $\langle s, d_j\rangle$ means d_j is well-aligned with s $\implies$ large explained chunk $\implies$ small $\varepsilon(j)$.
- We want the atom j^* with the smallest $\varepsilon(j)$.

11.4 Idea 3 — Why is $\varepsilon(j)$ Always Strictly Positive?

- This result is critical: $\varepsilon(j) > 0$ always. This comes from the **Cauchy-Schwarz inequality**:

$$\langle s, d_j\rangle^2 \leq \|s\|_2^2 \cdot \|d_j\|_2^2 \tag{22}$$

- Divide both sides by $\|d_j\|_2^2$:
$$\frac{\langle s, d_j\rangle^2}{\|d_j\|_2^2} \leq \|s\|_2^2 \tag{23}$$
- Therefore:
$$\varepsilon(j) = \|s\|_2^2 - \frac{\langle s, d_j\rangle^2}{\|d_j\|_2^2} \geq 0 \tag{24}$$
- Equality would only hold if s were perfectly parallel to d_j, which is the exception, not the rule. In general $\varepsilon(j) > 0$, meaning no single atom perfectly explains s. This is exactly why we need more than one atom — and why the algorithm needs to iterate.

12 Matching Pursuit Algorithm

12.1 Greedy Motivation

- **The Problem:** Solve (P_0^c): find $\hat{x} = \arg\min \|x\|_0$ subject to $Dx = s$.
- **The Challenge:** The brute-force approach is combinatorially intractable.
- **The Solution:** A **greedy algorithm** tackles this by making the locally best choice at each step. While there is no strict guarantee it finds the global minimum, in practice, it works remarkably well.
- **Algorithm Name:** The specific greedy algorithm introduced here is **Matching Pursuit (MP)**.

12.2 Algorithm Steps

- **Step 1 — Initialization:**
$$x^{(0)} = 0, \quad r^{(0)} = s, \quad \omega^{(0)} = \emptyset \tag{25}$$
$x^{(0)}$ indicates no atoms selected yet. The residual $r^{(0)}$ tracks "what of s still hasn't been explained." The active set ω records the list of atoms selected so far.
- **Step 2 — Sweep (Heart of MP):** For every atom j from 1 to N, compute the projection error:
$$\varepsilon(j) = \|r^{(i)}\|_2^2 - \frac{\langle r^{(i)}, d_j\rangle^2}{\|d_j\|_2^2} \tag{26}$$
Pick the atom that leaves the smallest residual:
$$j^* = \arg\min_j \varepsilon(j) \tag{27}$$
Note: We always project the remaining unexplained residual $r^{(i)}$, not the original signal s.
- **Step 3 — Update Active Set (ω):** Record the chosen atom:
$$\omega^{(i+1)} = \omega^{(i)} \cup \{j^*\} \tag{28}$$

- **Step 4 — Update Solution (x):** Calculate the projection coefficient z_{j^*} and update x:

$$z_{j^*} = \frac{\langle r^{(i)}, d_{j^*} \rangle}{\|d_{j^*}\|_2^2} \tag{29}$$

$$x^{(i+1)} = x^{(i)} + z_{j^*} \cdot e_{j^*} \tag{30}$$

  where e_{j^*} is the standard basis vector (a column of zeros with a single 1 at position j^*).

- **Step 5 — Update Residual (r):** The new residual shrinks by exactly the component of the chosen atom that was just explained away:

$$r^{(i+1)} = s - Dx^{(i+1)} = s - D(x^{(i)} + z_{j^*} e_{j^*}) = r^{(i)} - z_{j^*} d_{j^*} \tag{31}$$

- **Step 6 — Stopping Rule:** Halt the algorithm if either of these criteria is met:
  - $\|x^{(i)}\|_0 \geq L$ (Sparsity budget is exhausted; L atoms selected).
  - $\|r^{(i)}\|_2 < \varepsilon$ (Residual is small enough; $Dx \approx s$ within tolerance).

12.3 Geometric Intuition and Residual Decrease

- **Why $\varepsilon(j) > 0$ always:** Expanding the residual error gives $\varepsilon(j) = \|r - zd_j\|_2^2$. Substituting $z = \langle r, d_j \rangle / \|d_j\|_2^2$:

$$\varepsilon(j) = \|r\|_2^2 - \frac{\langle r, d_j \rangle^2}{\|d_j\|_2^2} \tag{32}$$

  By the Cauchy-Schwarz inequality, $\langle r, d_j \rangle^2 \leq \|r\|_2^2 \cdot \|d_j\|_2^2$. Dividing both sides by $\|d_j\|_2^2$:

$$\frac{\langle r, d_j \rangle^2}{\|d_j\|_2^2} \leq \|r\|_2^2 \implies \varepsilon(j) \geq 0 \tag{33}$$

  Strict inequality holds as long as r is not perfectly aligned with d_j (the generic case).

- **Geometry of Step 5:** $r^{(i)}$ is what is left of s after i iterations. We project it onto the best atom d_{j^*} and subtract that projection. The new residual $r^{(i+1)}$ is strictly perpendicular to d_{j^*} and strictly shorter than $r^{(i)}$. Thus, the algorithm **always makes progress**.

12.4 Key Insight: Why MP is Tractable

- **Complexity:** Instead of trying all $\binom{N}{k}$ subsets (which explodes combinatorially), MP scans all N atoms once per iteration and picks the single best one.
- Each sweep takes $O(NM)$ operations.
- After L iterations, the total computational cost is $O(LNM)$.
- **Conclusion:** The complexity is polynomial, not exponential.
- **The Trade-off:** Because MP is greedy, it may not find the globally sparsest $\hat{x}$. However, experimentally, it very often succeeds when the true solution is sufficiently sparse.

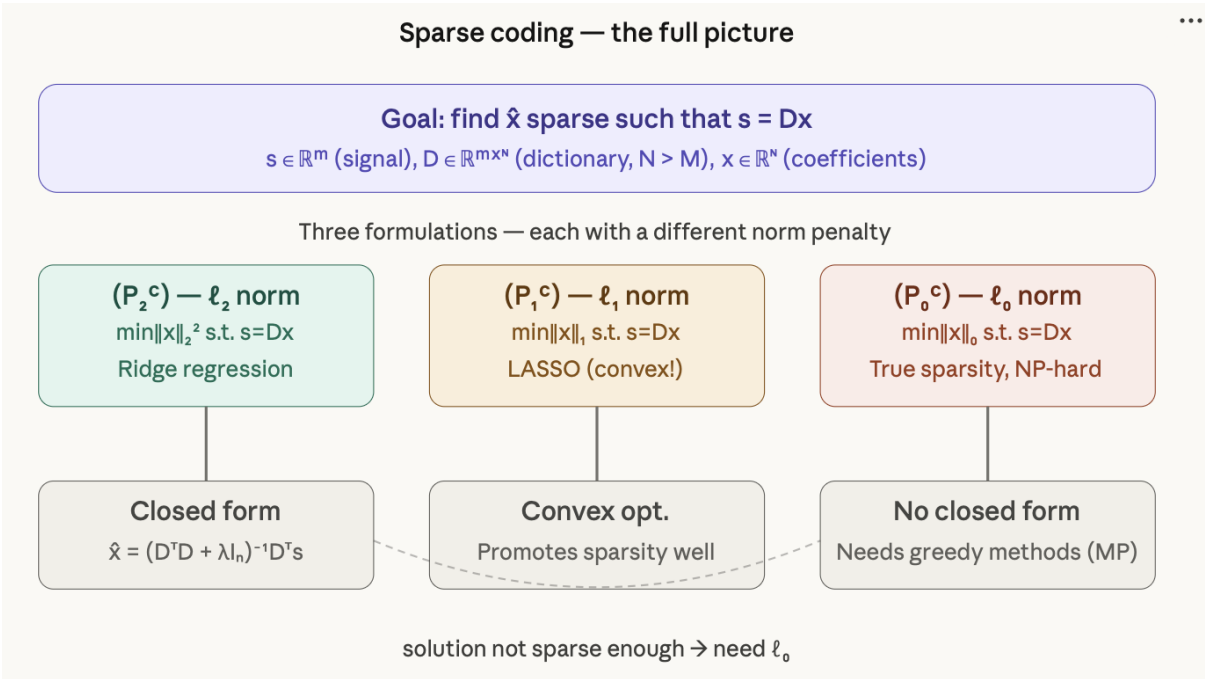

Figure 1: Sparse Encoding: Full Picture

13 Summary of the Narrative Arc

The course logic flows as follows:

- **Setup:** $s = Dx$ is underdetermined; we want a sparse x.
- **ℓ_2 Attempt (Ridge):** A closed-form exists, but the solution distributes energy everywhere (not sparse). It's the wrong tool.
- **Why ℓ_2 Fails:** We proved $D^\top D$ is SPSD, added λI to fix invertibility, but the resulting $\hat{x}$ has many small nonzero entries.
- **Norm Theory:** Defined proper norms and showed that a smaller p in ℓ_p geometrically promotes sparsity via the shape of the unit ball.
- **ℓ_0 is Ideal but Intractable:** ℓ_0 is not a proper norm (fails homogeneity), is discontinuous, and exact minimization leads to combinatorial explosion $O(N^k)$.
- **Next Step:** We need a heuristic/greedy approach $\rightarrow$ **Matching Pursuit**.